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Farnoush Zeidi
farnoush.zeidikolehparcheh@studenti.unimi.it
Matriculation number: 11327A
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Prof. Paolo Ceravolo

Abstract

In our initial phase, we provide an extensive introduction to the dataset, emphasizing its crucial role in unraveling the intricacies of the purchase order handling process. By categorizing the data into distinct flows namely, 3-way matching with invoice after goods receipt, 3-way matching with invoice before goods receipt, 2-way matching (no goods receipt needed), and consignment we aim to gain a comprehensive understanding of how each item aligns with specific procurement categories.

Transitioning to data preprocessing and filtering, we conduct a thorough examination of the dataset. Here, our focus lies on analyzing prevalent activities, variants, and start/end activity patterns. This in-depth analysis allows us to identify dominant variant patterns, which in turn aids in the filtration process. By removing noise, incomplete, or less frequent data, we enhance the overall quality and reliability of our dataset.

Moving forward, we delve into the realm of process mining algorithms to model the intricate purchase order handling system. Our exploration begins with the "Alpha miner" algorithm, where we meticulously apply it to our dataset and analyze its outcomes. Subsequently, we delve into the "Inductive miner" algorithm, deciphering its implications for process modeling. Additionally, we discuss the application and findings of the "Heuristic miner" algorithm. To provide a comprehensive overview, we then create process trees and present a comparative analysis using a heatmap chart, shedding light on the performance of each algorithm.

In the final phase of our study, we adopt a segmentation approach to further dissect the log file into distinct categories. This segmentation strategy is underpinned by a detailed rationale, which guides our subsequent operations. Applying filters to each segment, we meticulously evaluate their impact on data quality and the outcomes of our process discovery endeavors. Finally, we present a comparative analysis, scrutinizing the performance of each process discovery algorithm across the segmented data categories. This holistic approach offers nuanced insights into the efficacy of these algorithms under varied conditions, thus contributing significantly to our understanding of the purchase order handling process.

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1 Introduction

The Business Information Systems (BIS) project focuses on analyzing and optimizing the purchase order handling process within a large multinational company headquartered in The Netherlands, specializing in coatings and paints. With operations spanning across 60 subsidiaries, the company seeks to enhance compliance and efficiency within its procurement operations.

The dataset provided for analysis encompasses a comprehensive view of the purchase order handling process, detailing various scenarios encountered in procurement. Each purchase order, comprising one or more line items, presents distinct challenges related to compliance and matching criteria.

Four primary types of purchase order flows are identified within the dataset:

1-3-way matching, invoice after goods receipt: This scenario involves matching the value of goods receipt messages against invoices after goods are received. Compliance is indicated by specific flags, ensuring alignment between the recorded values and the initial creation of the item.

2-3-way matching, invoice before goods receipt: Here, purchase items necessitate goods receipt messages but allow invoices to be entered before goods receipt, subject to blocking until goods are received. Unblocking occurs either through user intervention or automated batch processes, with strict adherence to matching criteria upon goods receipt.

3-2-way matching (no goods receipt needed): Items in this category require invoice value alignment with the initial creation, without the need for separate goods receipt messages. Compliance is verified through specified flags, ensuring accuracy in financial transactions.

4-Consignment: Items under this classification follow a separate process without invoices at the purchase order level. While goods receipt indicators are present, invoices are not expected due to the consignment nature of transactions.

Beyond these categories, the dataset presents complexities such as multiple goods receipt messages and corresponding invoices for each purchase item. Variations in the number of messages and invoices contribute to the intricacies of ensuring compliance and financial accuracy.

The anonymized dataset adheres to IEEE-XES standards, providing detailed attributes for each purchase item. These attributes include purchasing document and item IDs, item types, flags indicating invoicing and matching requirements, vendor information, and various textual classifications related to spending areas and categories.

With a total of 76,349 purchase documents containing 251,734 items and 1,595,923 events recorded, the dataset offers rich insights into the procurement processes undertaken by 627 users across the organization.

Through comprehensive analysis of this dataset, the BIS project aims to identify inefficiencies, enhance compliance measures, and optimize the purchase order handling process to drive greater operational effectiveness and financial accuracy across the multinational company's subsidiaries.

2 Preprocessing

Preprocessing the log file is crucial to understanding its structure and reducing noise. This involves several key steps: first, we analyze the frequency of variants, identifying that the top 500 variants account for 81% of the data. Next, we examine the frequency of activities, noting that 10 out of 42 activities make up 96% of the log file. We also investigate the sequence of variants and the start and end activities to detect anomalies. Furthermore, we filter cases based on their lifecycle duration, removing those with abnormally long durations. Lastly, we assess the frequency distribution across the four predefined categories to understand their prevalence.

2.1 Exploring Log File

The goal of this subsection is to thoroughly understand the log file's structure and contents. Key analyses include:

- **Variant Frequency:** Identify the most frequent variants and their cumulative frequency.
- **Activity Frequency:** Determine the prevalence of activities within the log file.
- **Variant Sequence Frequency:** Use Sunburst charts to analyze variant sequences.
- **Start and End Analysis:** Detect anomalies in start and end activities.
- **Case Duration:** Visualize case duration to filter out abnormal cases.
- **Category Frequency:** Analyze the frequency distribution across the four categories.

Variant Frequency Analysis

In the context of process mining, a variant refers to a unique sequence of activities that occur within a single case or instance in the log file. Each case represents a single execution of the process, and the sequence of activities (events) that occur during this case is termed as a variant. Identifying and analyzing the frequency of these variants is crucial for understanding the common patterns and behaviors in the process.

Goal:

The primary goal of analyzing variant frequency is to identify which sequences of activities are most common within the dataset. By focusing on the most frequent variants, we can gain insights into:

- Typical pathways through the process.
- Recurring patterns.
- Core structure of the process.

Method:

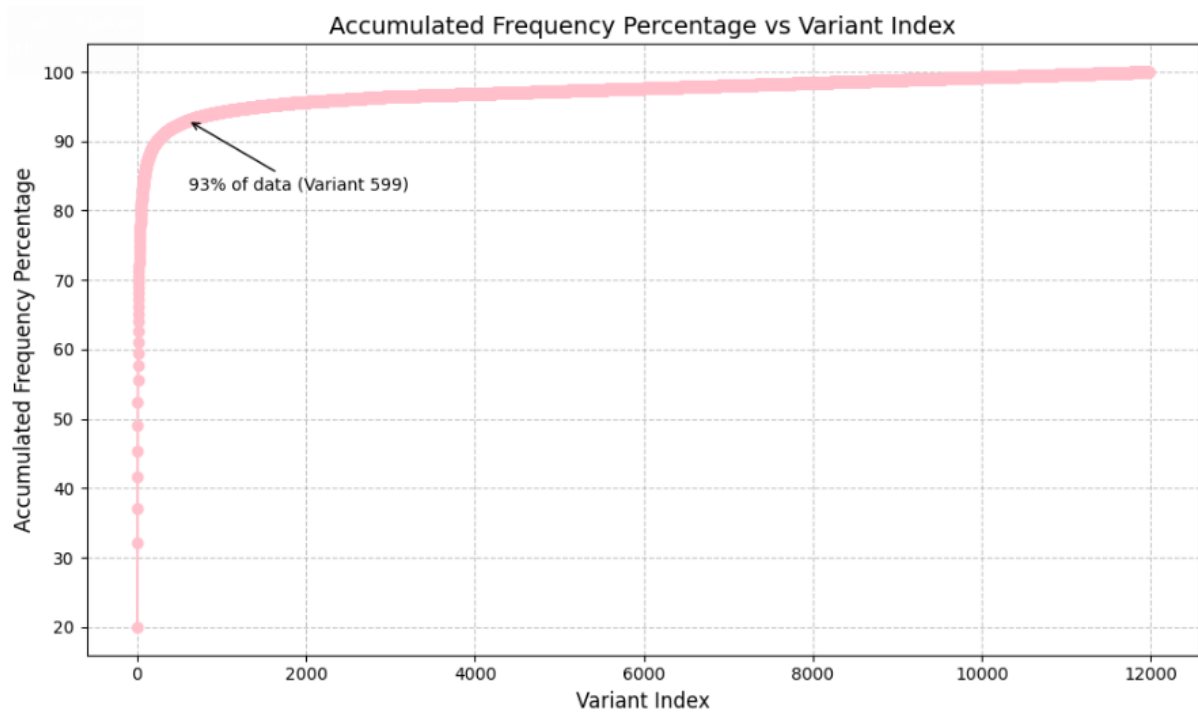
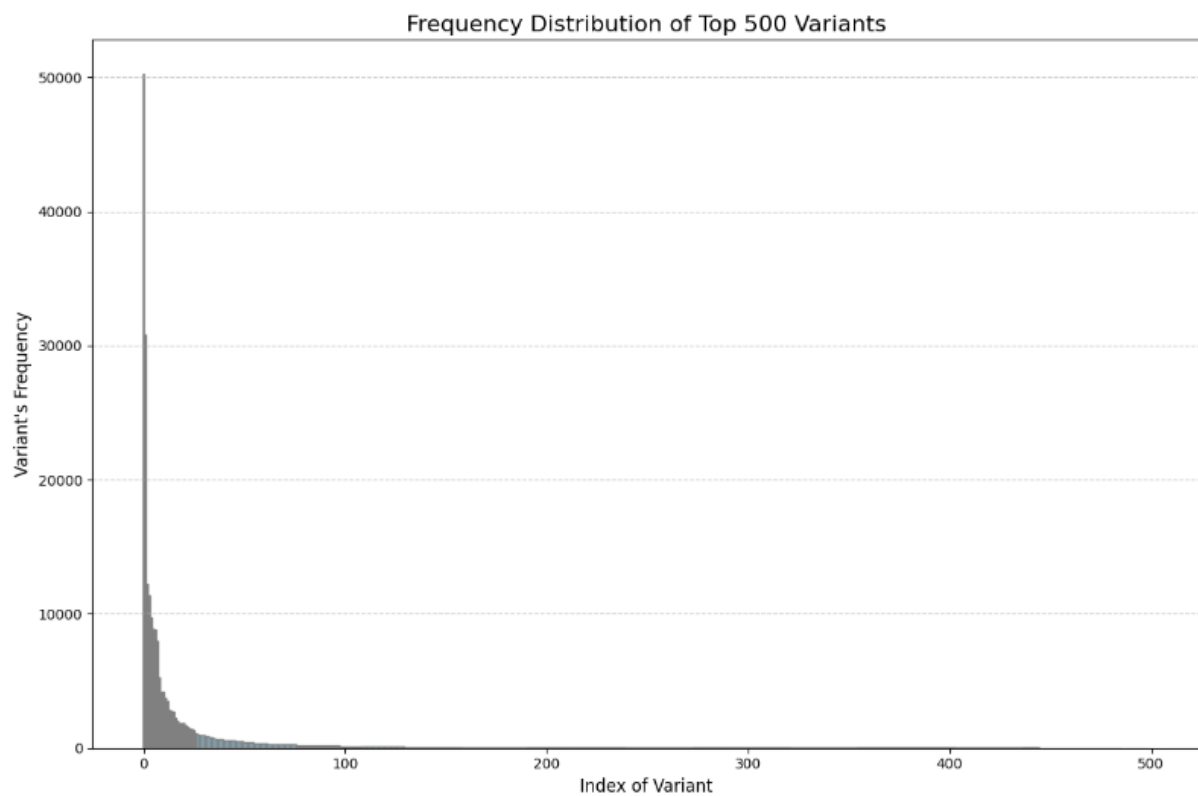
There are three main steps involved in variant frequency analysis:

- **Identify Variants:** Extract and list all unique variants from the log file. Each variant represents a different sequence of activities.
- **Calculate Frequency:** Count the number of occurrences for each variant. This helps determine how often each unique sequence appears in the dataset.
- **Cumulative Frequency:** Calculate the cumulative frequency by adding the frequencies of the most frequent variants sequentially. This cumulative measure helps understand the overall coverage of the most common sequences in the dataset.

Analysis:

Here's how we analyze the results:

- **Top Variants:** Focus on the top variants that account for the majority of the cases (e.g., top 500 variants covering 93% of the dataset).
- **Variant Distribution:** Analyze how the frequencies are distributed among the variants (typically a small number of variants cover a large portion, while many appear infrequently).
- **Visualization:** Use visual tools like bar charts the frequency of each variant and their cumulative distribution. This visualization helps identify dominant patterns in the process.



Activity Frequency Analysis

In process mining, activities represent the individual steps or actions that occur within a case in the log file. Analyzing the frequency of activities helps us understand which actions are most common, providing insights into the core components of the process.

Goal:

The primary goal of analyzing activity frequency is to identify which activities are most prevalent in the process. By focusing on the most frequent activities, we can determine the key actions that drive the process and recognize patterns that might indicate areas for improvement.

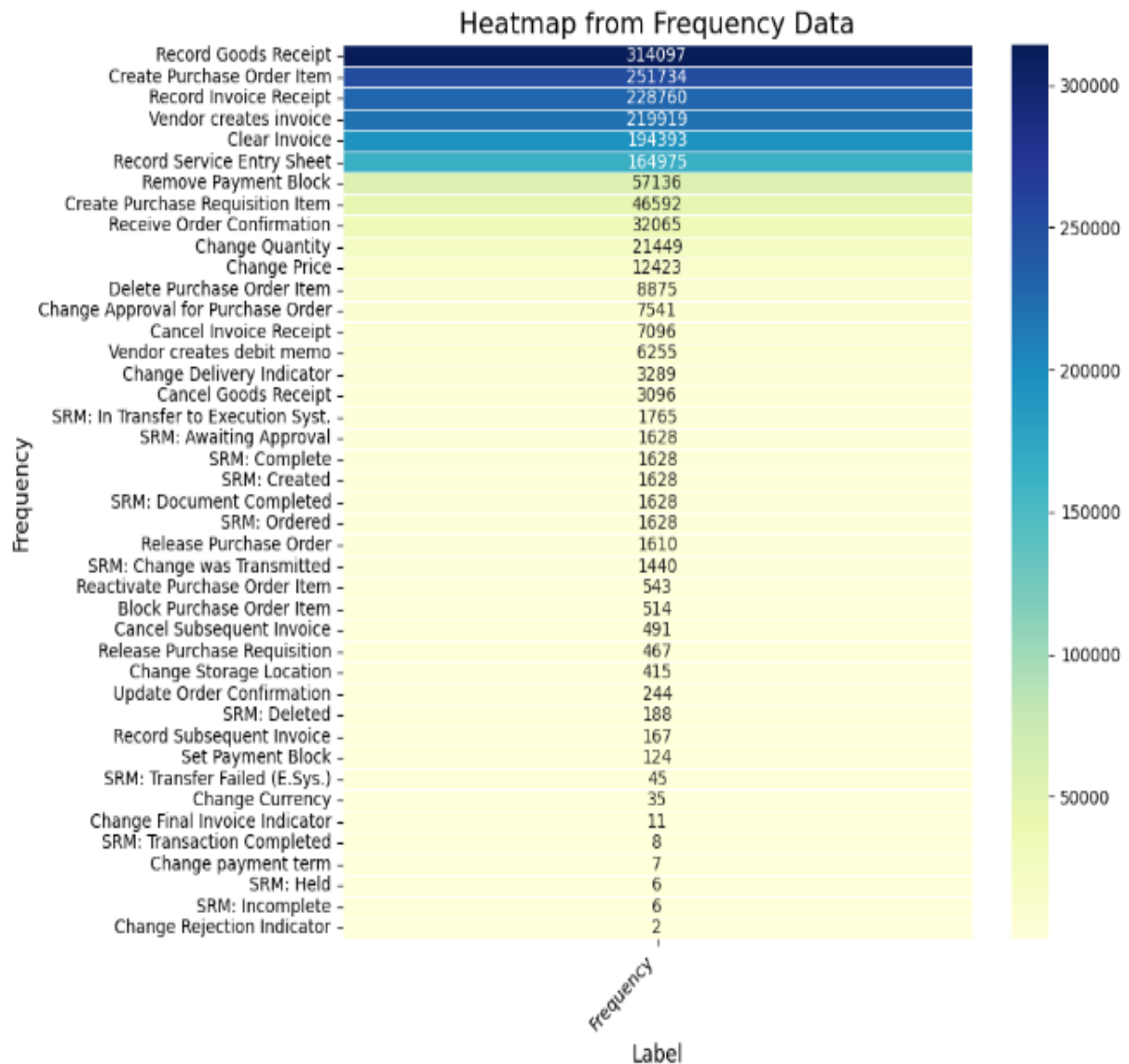
Method:

- **Identify Activities:** Extract and list all unique activities from the log file. Each activity corresponds to a distinct step within the process.
- **Calculate Frequency:** Count the number of occurrences for each activity. This helps in determining how often each activity appears in the dataset.

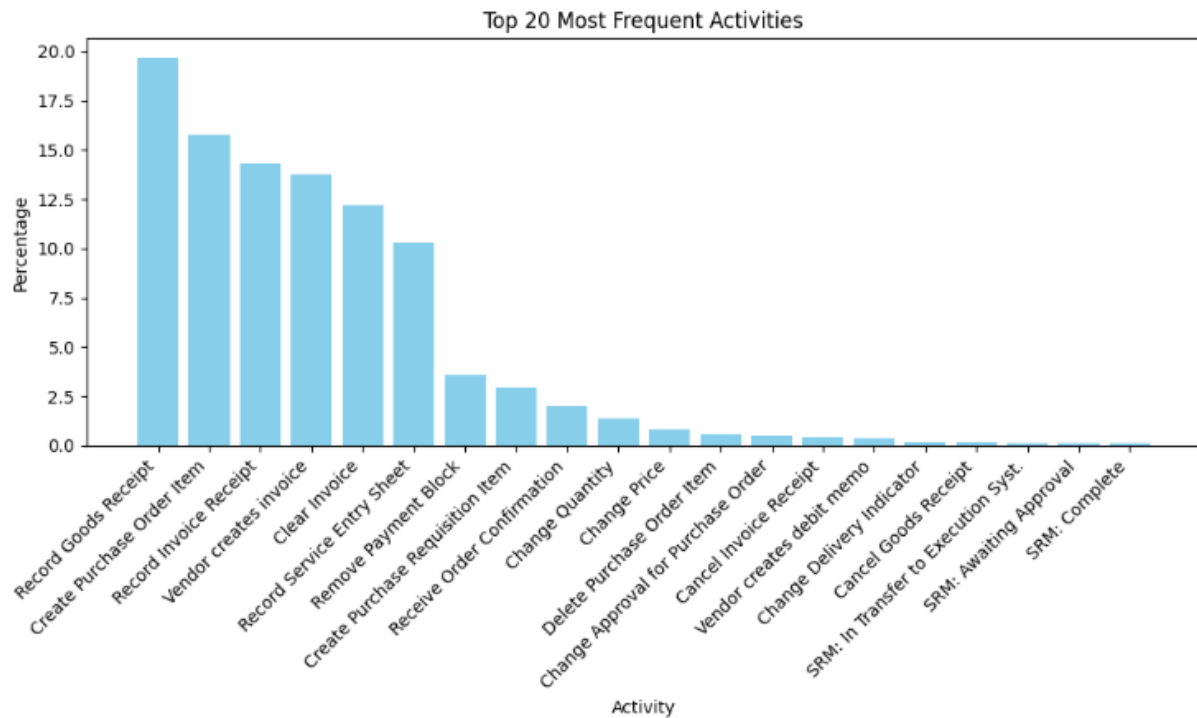
Analysis:

- **Top Activities:** Focus on the top activities that account for the majority of occurrences. For instance, a small number of activities might constitute the bulk of the actions recorded in the log file.
- **Activity Distribution:** Analyze how the frequencies are distributed among the activities. Typically, a few activities dominate the process, while others occur less frequently.
- **Visualization:** Use visual tools such as bar charts to represent the frequency of each activity and their cumulative distribution. This visualization aids in quickly identifying the dominant actions in the process.

To further enhance the visualization of activity frequencies, heatmaps can be used to illustrate the distribution of all activity occurrences across different cases. This approach provides a detailed view of where and how frequently activities are taking place within the process, highlighting patterns and potential areas for optimization.



A simple bar chart showing the frequency of the top 20 activities. This chart will clearly illustrate which activities are most prevalent in the process, making it easier to identify key actions and areas for improvement.



Below is a table showing the top 10 activities along with their frequencies and percentages:

Activity Name	Frequency	Relative Frequency (%)
Record Goods Receipt	314,097	19.68
Create Purchase Order Item	251,734	15.77
Record Invoice Receipt	228,760	14.33
Vendor Creates Invoice	219,919	13.78
Clear Invoice	194,393	12.18
Record Service Entry Sheet	164,975	10.34
Remove Payment Block	57,136	3.58
Create Purchase Requisition Item	46,592	2.92
Receive Order Confirmation	32,065	2.01
Change Quantity	21,449	1.34

This table highlights the most common activities, their frequency of occurrence, and their contribution to the overall activity distribution in the log file.



Variant Sequence Frequency Analysis

Understanding variant sequences is essential for gaining insights into the flow and structure of processes. A variant sequence represents the ordered occurrence of activities within a case or instance in the log file. Analyzing the frequency of variant sequences helps identify the most common paths followed in the process execution.

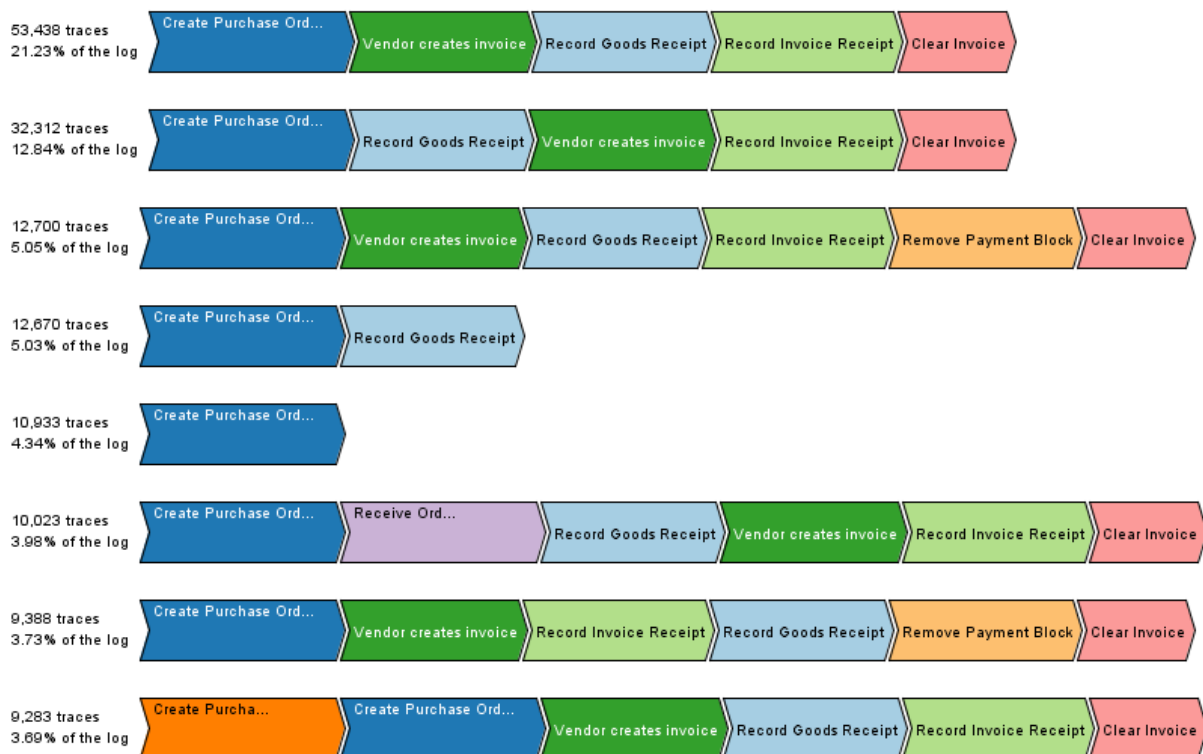
Goal:

The primary objective of analyzing variant sequence frequency is to uncover the prevalent sequences of activities within the dataset. By focusing on the frequency of these sequences, we aim to achieve the following:

- Identify the most common pathways or flows through the process.
- Detect recurring sequences of activities.
- Gain insights into the core structure of the process based on frequently occurring sequences.

ProM Application for Variant Sequence Analysis:

To perform a detailed variant sequence frequency analysis, we utilize the ProM application. ProM is a powerful tool for process mining, which provides various plugins and functionalities to analyze event logs.



In the figure, you can see the top 8 variant sequences based on their frequency within the dataset. This approach helps us focus on significant process paths and gain valuable insights for process improvement.

Start and End Analysis

Start and end activities refer to the initial and final actions that occur within a case in the log file. Analyzing these activities helps in understanding the entry and exit points of the process, identifying any anomalies, and ensuring process consistency.

Goal:

The primary goal of start and end analysis is to identify the most common starting and ending activities within the dataset. This analysis helps in detecting any unusual or unexpected activities that might indicate anomalies or deviations from the standard process flow.

Method:

- **Sequence Extraction:** Extract and compile all unique variant sequences from the log file. Each sequence represents a distinct ordered occurrence of activities within a case. This step identifies the different paths taken through the process.
- **Frequency Calculation:** Count the occurrences of each variant sequence within the dataset to determine their frequency of occurrence. This helps us understand how often each specific sequence of activities appears in the log data.

Analysis:

- **Top Start and End Activities:** Focus on the top start and end activities that account for the majority of occurrences. For instance, a small number of activities might constitute the bulk of the start and end points in the log file.

This table presents the starting activities along with their frequencies.

Action	Occurrences
Create Purchase Order Item	199867
Create Purchase Requisition Item	46526
Vendor creates invoice	3457
SRM: Created	1360
Change Approval for Purchase Order	378
Vendor creates debit memo	122
Release Purchase Order	22
Change Currency	2

This table illustrates Top 10 End activities and their corresponding frequencies. The total number of End activities is 32.

Action	Occurrences
Clear Invoice	181328
Record Invoice Receipt	23091
Record Goods Receipt	22776
Delete Purchase Order Item	8123
Remove Payment Block	5397
Create Purchase Order Item	4447
Cancel Invoice Receipt	1348
Change Approval for Purchase Order	1346
Record Service Entry Sheet	921
Change Delivery Indicator	718

Category Frequency Analysis

Analyzing category frequency involves examining the occurrence rates of different categories within a dataset. This analysis aids in understanding the distribution of categories, identifying patterns or trends, and informing decision-making processes.

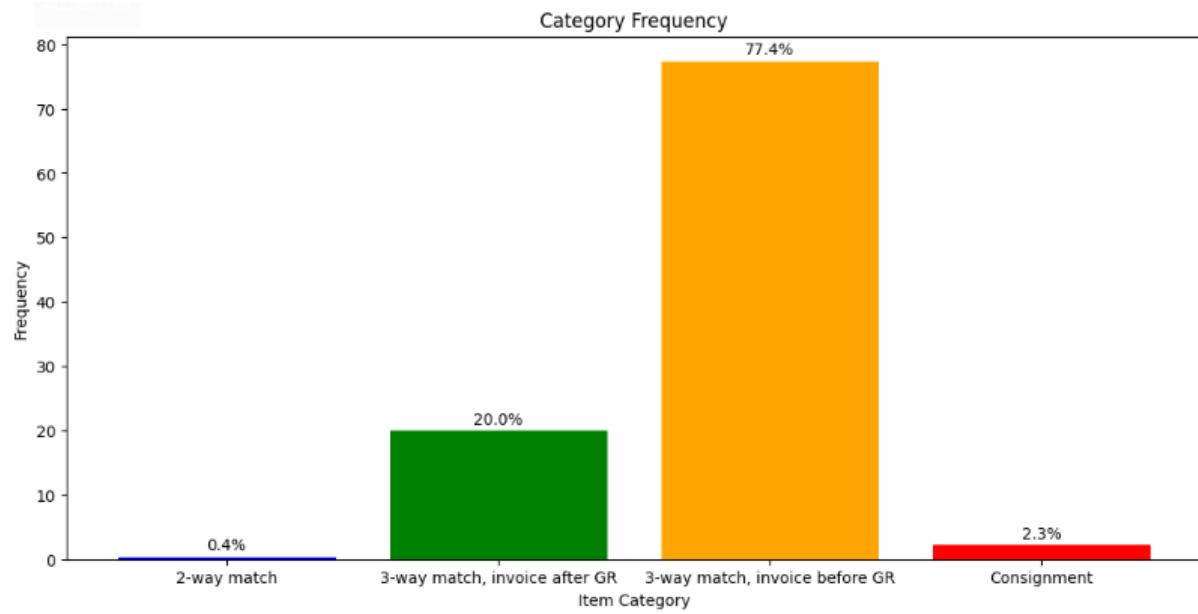
Goal:

The primary goal of analyzing category frequency is to identify the most common categories within the dataset and gain insights into their distribution. This analysis helps in detecting any irregularities or deviations from expected category distributions.

Method:

- **Category Extraction:** Extract and compile all unique categories present in the dataset. Each category represents a distinct classification or grouping of data points.
- **Frequency Calculation:** Calculate the frequency of each category within the dataset to determine its occurrence rate. This step provides quantitative information about how often each category appears in the dataset.
- **Comparative Analysis:** Conduct a comparative analysis to compare the frequencies of different categories. This comparison helps in identifying dominant categories and understanding their relative importance within the dataset.

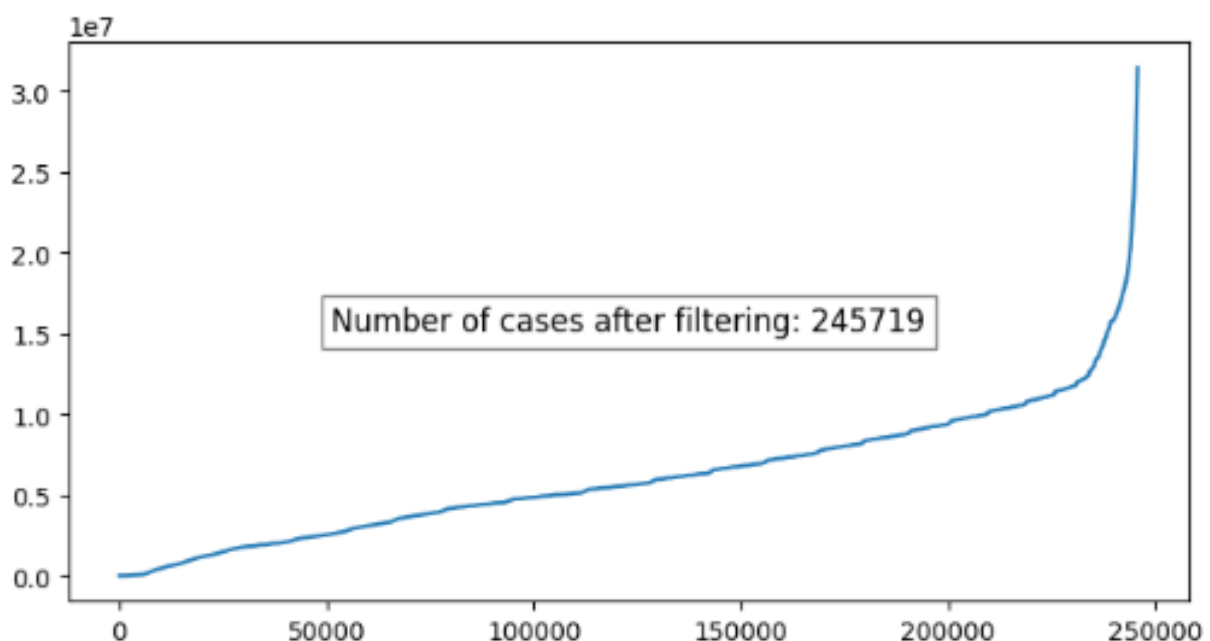
The frequency of each category is depicted in the following chart:



2.2 Filtering and Processing Log File

Case Performance

Our initial focus is on the completion duration, with the goal of identifying any irregularities in the recorded event logs. These logs show durations ranging from 0 seconds to an unlikely 2,217,935,340 seconds, equivalent to approximately 70 years, which suggests possible inaccuracies in data recording. As a result, we eliminate cases with durations exceeding reasonable limits and retain only those lasting between 10 minutes and 365 days. Subsequently, we create a chart depicting case durations to visualize the effects of this filtering process. Specifically, this filtering process leads to the removal of 6,005 cases from the log file, resulting in a more balanced representation of case durations.



Top Variant Filter

The second filter we use is the "top-k variant" filter. We have identified that the top 599 variants account for 93% of the frequency in our dataset. By applying this filter, we reduce the number of cases from 245,719 to 229,706, eliminating 15,336 cases that fall outside the top 599 variants. This helps in focusing on the most common variants, which likely represent the most significant and meaningful process paths.

Start and End Activity Filter

The third filter focuses on the start activities of the cases. We only retain records that start with specific activities based on our understanding of activity frequency.

The start activities we consider are:

- "Create Purchase Order Item"
- "Create Purchase Requisition Item"
- "Vendor creates invoice"
- "SRM: Created"
- "Vendor creates debit memo"

Additionally, we consider the following end activities:

- "Clear Invoice"
- "Record Invoice Receipt"
- "Record Goods Receipt"
- "Delete Purchase Order Item"
- "Remove Payment Block"
- "Cancel Invoice Receipt"
- "Change Approval for Purchase Order"
- "Change Delivery Indicator"
- "Cancel Goods Receipt"
- "Receive Order Confirmation"
- "Block Purchase Order Item"
- "Set Payment Block"
- "SRM: Change was Transmitted"
- "SRM: Transfer Failed (E.Sys.)"
- "SRM: In Transfer to Execution Syst."
- "Cancel Subsequent Invoice"
- "SRM: Deleted"
- "SRM: Transaction Completed"
- "Update Order Confirmation"

This filter ensures that we only include cases that begin with the specified start activities and end with the specified end activities, further refining our dataset to include the most relevant cases. The remaining number of cases after applying these filters is 227100.

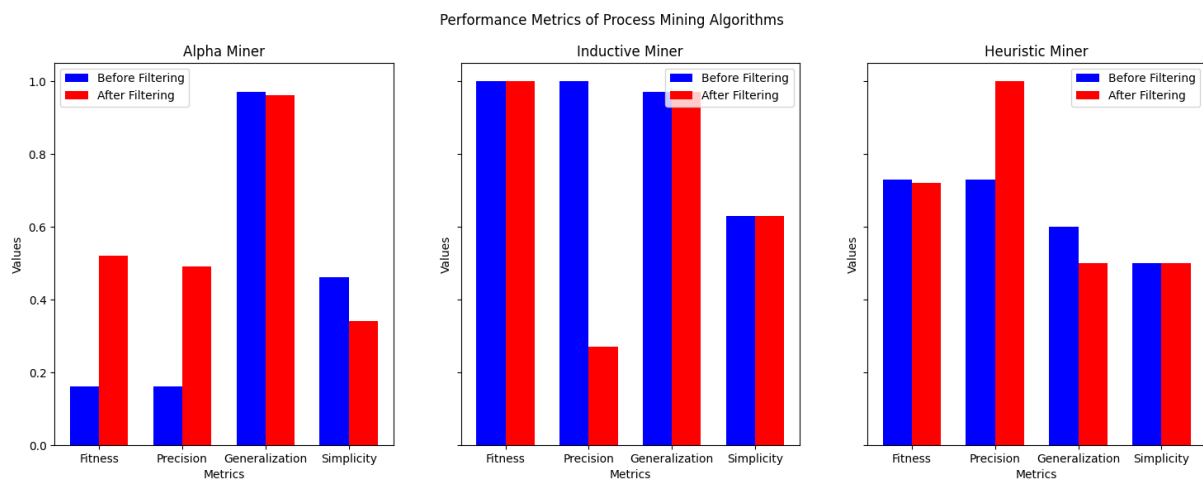
Activity Rework

Another crucial aspect of our analysis is the presence of repetitive activities within cases. We have implemented a filtering method to identify and exclude cases with excessive activity rework. Specifically, cases with more than 2 repetitions of the following activities are filtered out:

- Clear Invoice
- Record Goods Receipt
- Record Invoice Receipt

This filtering process resulted in the removal of 6749 cases from our dataset. As a result, the remaining dataset consists of 220351 cases, ensuring a focus on cases with a reasonable level of activity repetition.

While we will delve into process discovery in subsequent sections, it is pertinent to highlight the impact of this filtering method. The charts compare the performance metrics of Alpha Miner, Inductive Miner, and Heuristic Miner before (blue) and after (red) applying filters and removing 31,383 cases. The red bars indicate improved metrics post-filtering, showcasing the refinement achieved.



3 Process Discovery

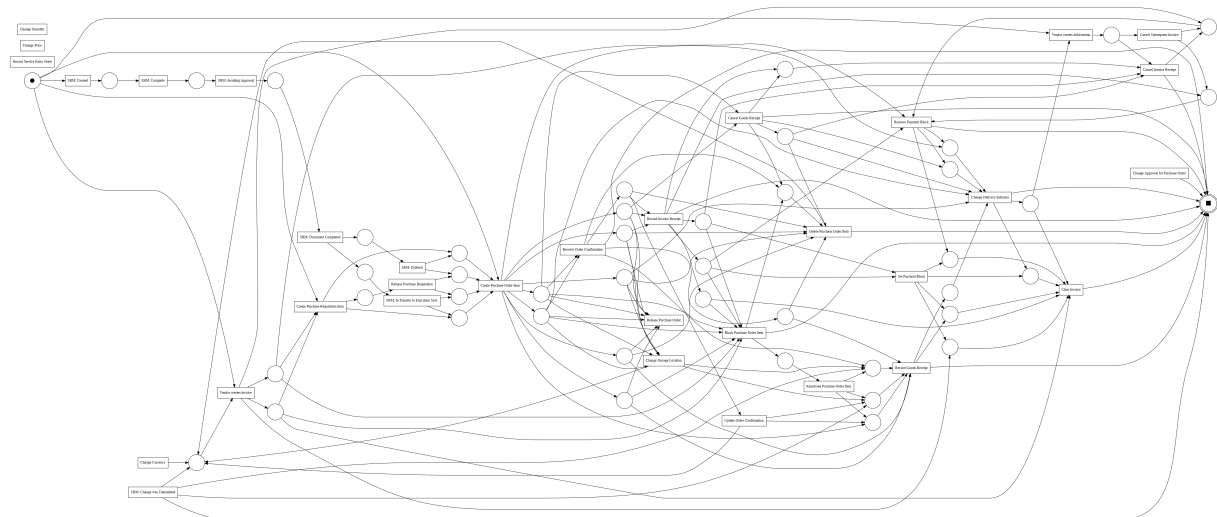
In this section, we aim to apply various process discovery algorithms and compare them across four key criteria: generalization, fitness, precision, and simplicity.

3.1 Alpha Miner

The first algorithm we utilize is Alpha Miner. Alpha Miner is a foundational process mining algorithm used to discover process models from event logs. This algorithm focuses on identifying causal dependencies between activities within a process. By analyzing the sequence in which events occur, Alpha Miner infers the relationships between them, such as the precedence or concurrency of activities.

Alpha Miner operates on the principle of the "directly follows" relation, which captures the immediate succession of events in the log. Using this information, it constructs a process model, typically represented as a Petri-net or a process flowchart. This model aims to reflect the underlying process structure by highlighting the direct connections and dependencies between activities.

When applying Alpha Miner to our log file, we generate the following Petri-net:

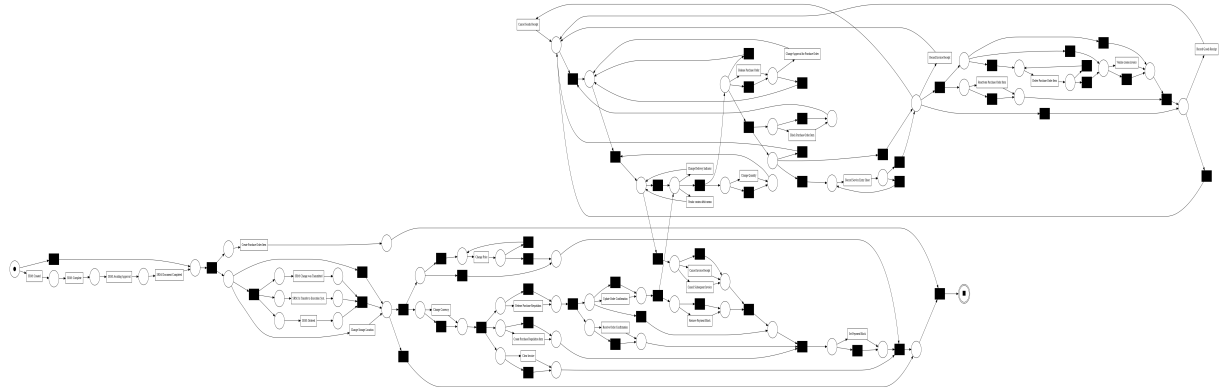


Despite its theoretical foundations, the resulting model from Alpha Miner did not effectively capture the process. The generated Petri-net lacks a clear path from start to end, indicating that the algorithm struggled to model the complexity of the process accurately. This outcome suggests potential limitations in handling logs with intricate or less structured processes.

3.2 Inductive Miner

The subsequent algorithm we consider is the Inductive Miner. This process mining algorithm is designed to discover process models from event logs, focusing on inferring the structure of a process model, including possible choices and loops. Unlike the Alpha Miner, the Inductive Miner employs a bottom-up approach, starting with individual activities and gradually constructing a model by merging similar behaviors.

This method effectively handles more complex process behaviors, such as exclusive choices and parallelism, making it suitable for a broader range of process models. After applying this algorithm to the filtered log file and generating a Petri net, the resulting output is as follows:

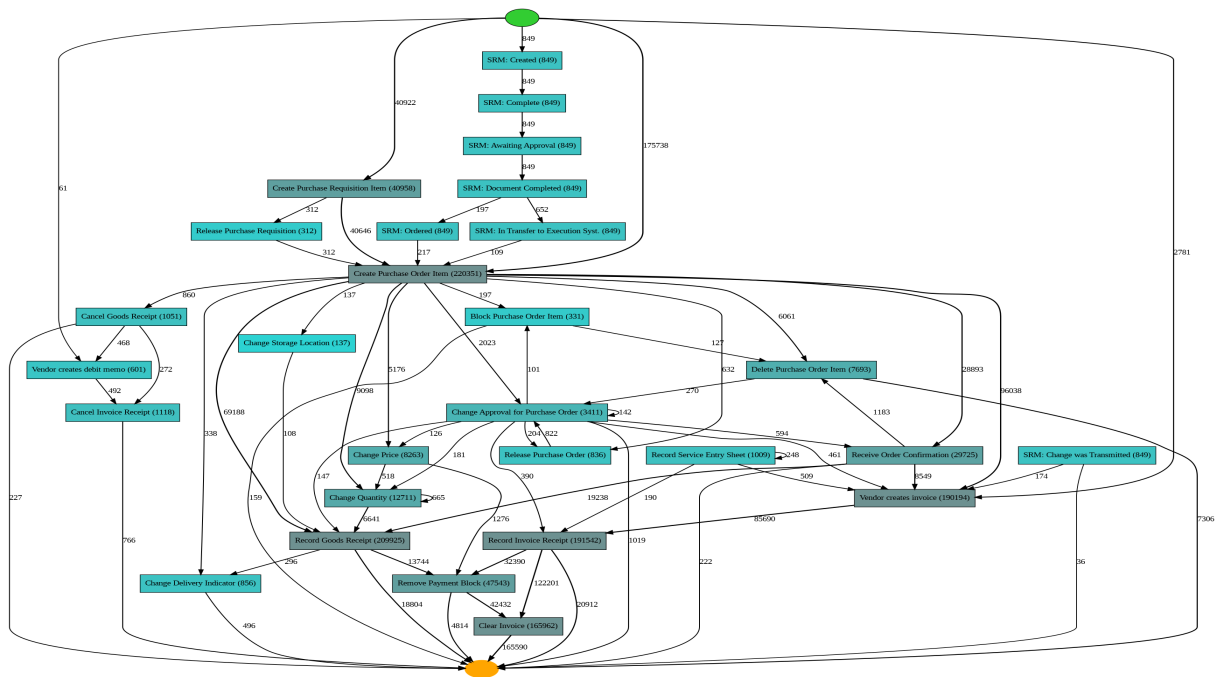
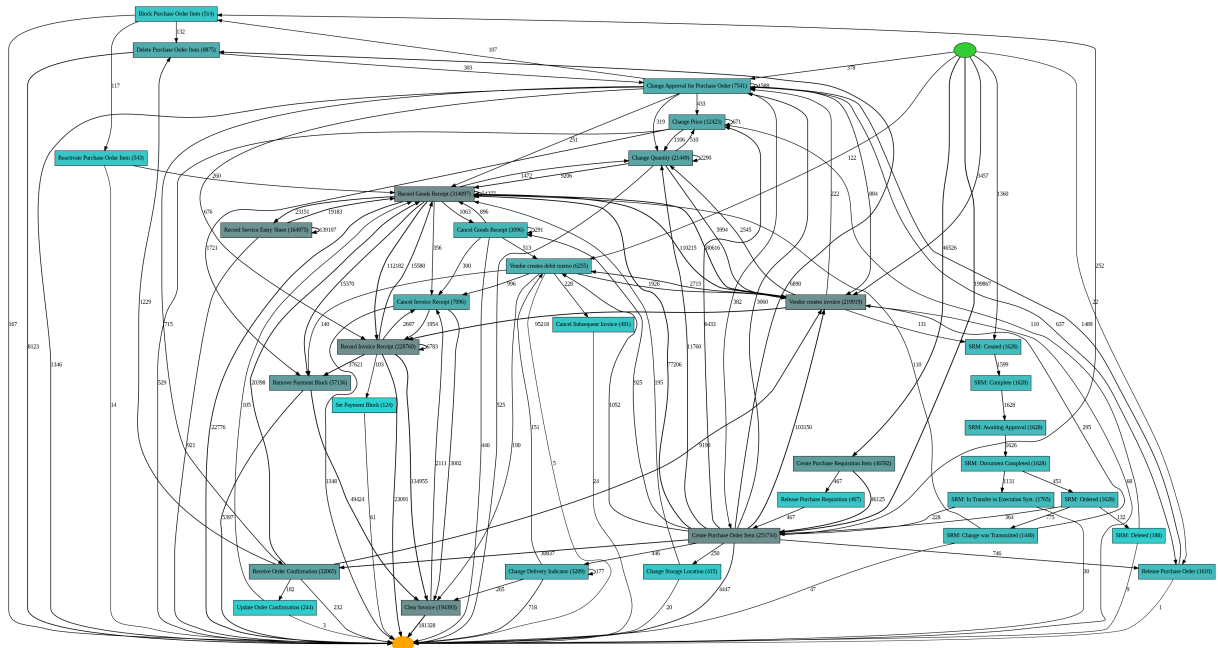


The output appears to be more precise than that of the Alpha Miner. The paths are clearly defined, and all activities have distinct start and end points. Additionally, the Inductive Miner demonstrates enhanced capability in modeling complex process structures, providing a comprehensive and accurate representation of the underlying processes. This precision facilitates better analysis and understanding of the process flows, contributing to improved process management and optimization.

3.3 Heuristic Miner

The Heuristic Miner is a process mining algorithm that infers process models from event logs by applying heuristics to identify patterns and dependencies. It considers factors such as frequency, exclusive choices, and concurrency. Compared to the Alpha Miner, the Heuristic Miner can handle noisy event logs and capture more complex process behaviors. Unlike the Inductive Miner, which uses a bottom-up approach, the Heuristic Miner focuses on heuristics rather than structural inference methods.

This version emphasizes that the Petri net serves as a clear representation of how filtering has improved the quality and usefulness of the log data for analysis. It highlights the direct benefits of the filtering process without explicitly stating it as a positive outcome, allowing the visualization itself to demonstrate the effectiveness.

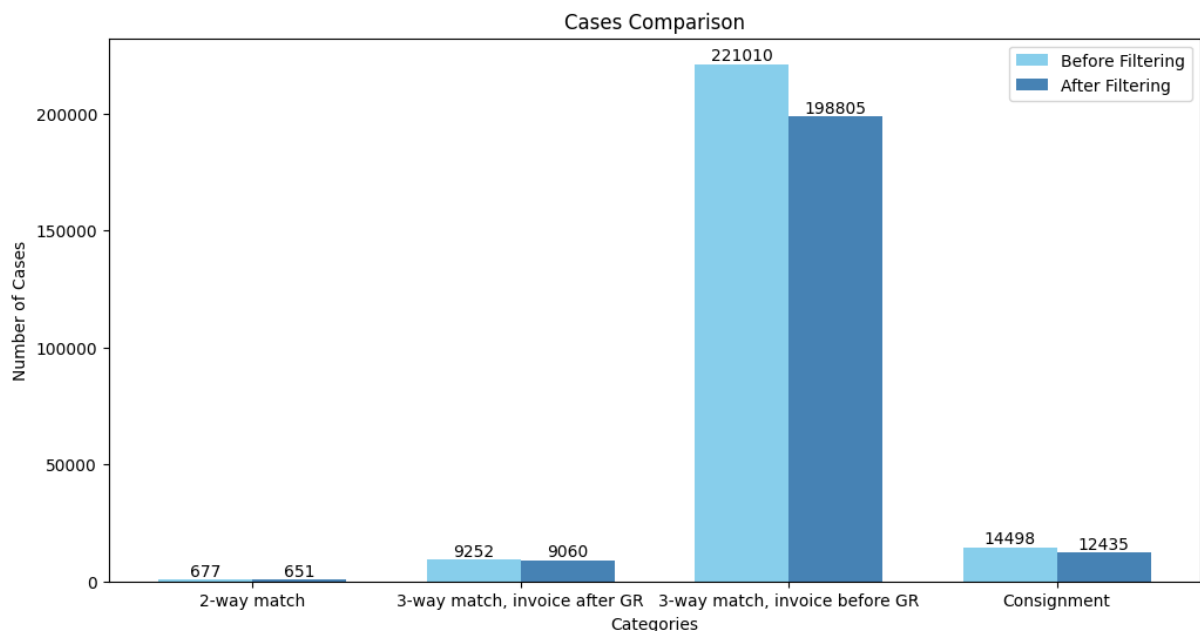


4 Data Segmentation

In this section, we categorize the filtered data into four categories:

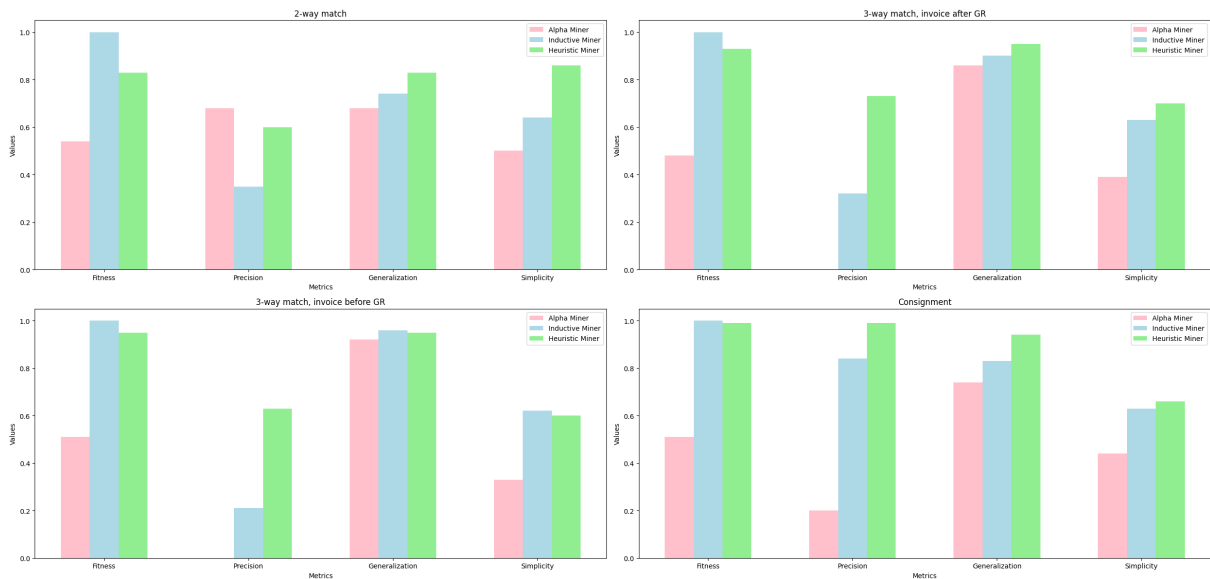
- 2-way match
- 3-way match Invoice after GR
- 3-way match invoice before GR
- Consignment

The light blue bars represent the number of cases in each segment before filtering, while the navy bars show the remaining cases after filtering. This comparison highlights the impact of the filtering process on each category, demonstrating the reduction in cases and the refinement achieved for subsequent analysis.



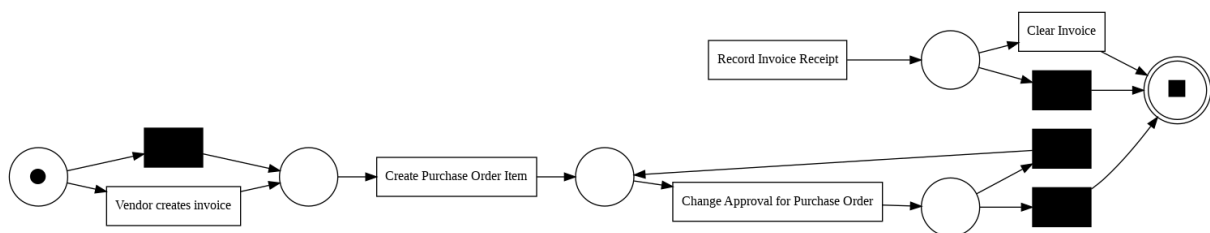
4.1 Process Discovery

To understand how good each model is, we consider four aspects: recall, precision, generalization, and simplicity.

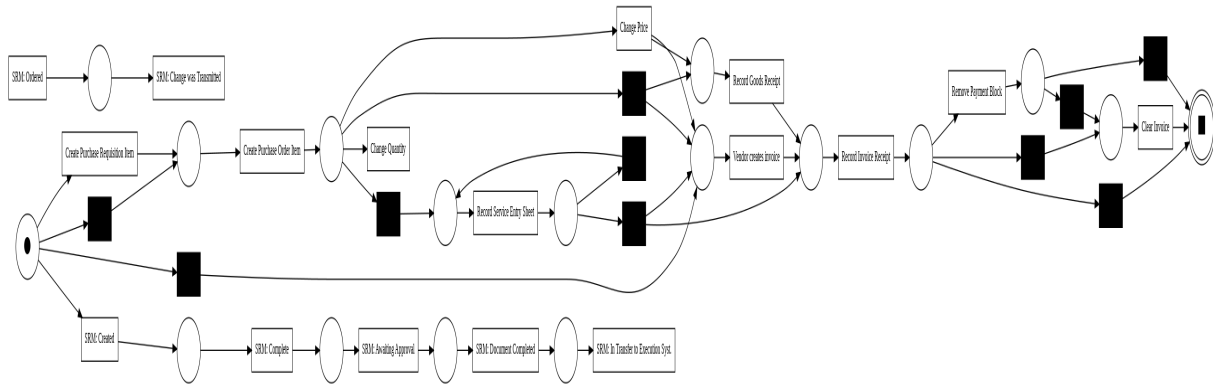


In these four bar charts, it is evident that among the three different mining algorithms employed, the Alpha Miner algorithm consistently exhibits the lowest performance across various metrics. Conversely, the Inductive Miner algorithm demonstrates notable performance advantages in certain aspects compared to the others. In contrast, the Heuristic Miner algorithm consistently displays uniform performance across all evaluated metrics, consistently achieving high performance levels. The Heuristic Miner algorithm proves to be the optimal choice across all four categories due to its consistent and superior performance compared to both the Alpha Miner and Inductive Miner algorithms.

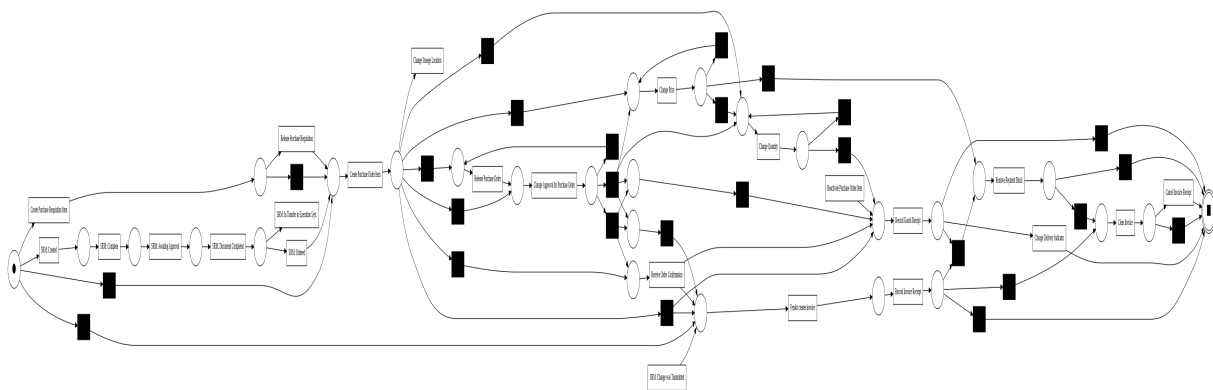
Detailed heuristic models for each segment are clearly illustrated in the following pages.



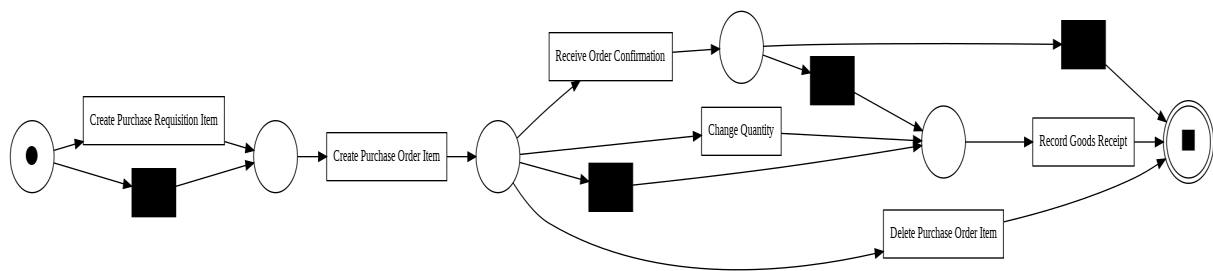
2-way match



3-way match, invoice before GR



3-way match, invoice after GR



Consignment

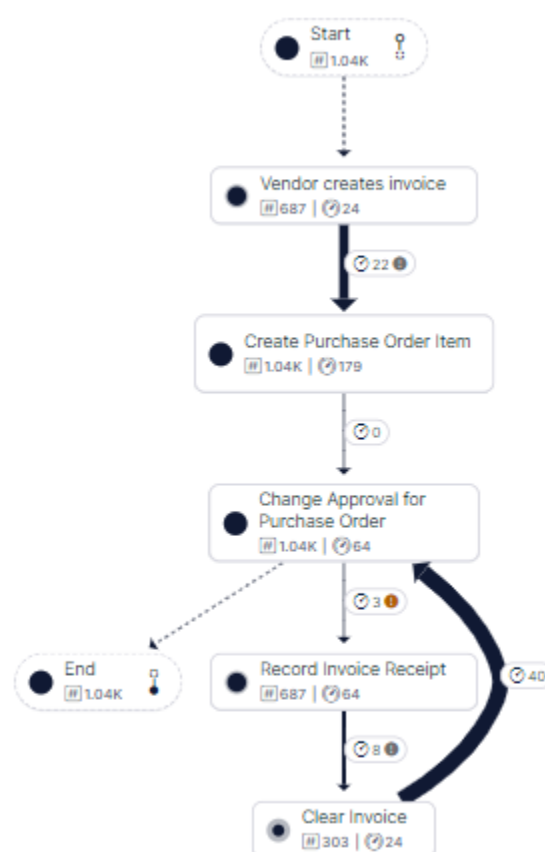
5 Celonis

After all the processing we have done in our data using Python and the PM4Py library, in this section, we will use the Celonis platform to explore our dataset further. First, we divide our data based on the segments we identified in the previous step:

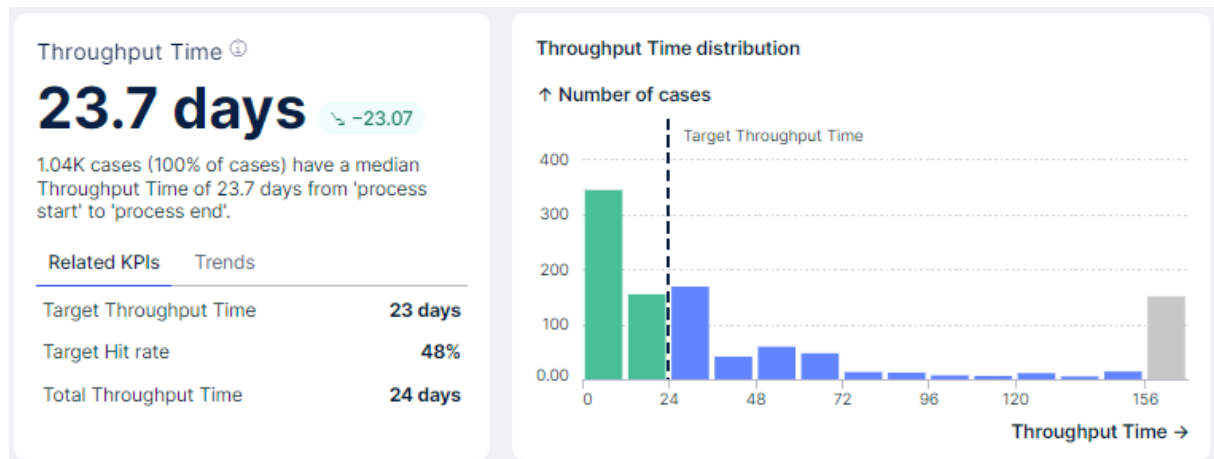
- 2-way match
- 3-way match Invoice after GR
- 3-way match Invoice before GR
- Consignment

For each segment, we will focus on the top activities to illustrate bottlenecks and identify the best filters to represent our processes accurately.

5.1 2-way match



2 way match model



Throughput Time

The first figure shows the throughput time distribution for the "2-way match" process. The median throughput time is 23.7 days. This analysis includes all 1.04K cases, and the distribution demonstrates how many cases fall within specific throughput time ranges. The target hit rate is 48%, indicating that just under half of the cases meet the target throughput time. The light blue bars represent cases that met the target, while the navy bars represent cases that exceeded it. The distribution reveals a significant number of cases completing within the target time, but there are still outliers extending up to 156 days, suggesting there is room for process improvement.

The second figure presents the process explorer for the "2-way match" segment. The process begins with 'Vendor creates invoice' and proceeds through key activities to 'Clear Invoice'. The top activities in this flow are:

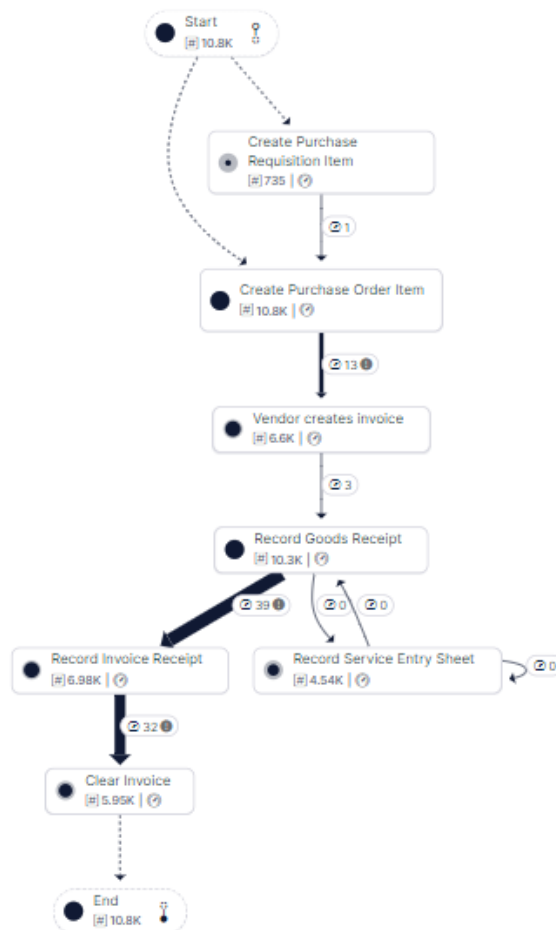
- **Vendor creates invoice:** This initial step involves creating the invoice and has 687 cases with a throughput time of 24 days.
- **Create Purchase Order Item:** This is the most frequent activity with 1.04K cases and a throughput time of 179 days, indicating a potential bottleneck.
- **Change Approval for Purchase Order:** This activity also involves 1.04K cases with a throughput time of 64 days, showing another significant process step.
- **Record Invoice Receipt:** With 687 cases and a throughput time of 64 days, this activity is critical in the process flow.
- **Clear Invoice:** This final step has 303 cases with a throughput time of 24 days, completing the process.

The thick arrow from 'Change Approval for Purchase Order' to 'Record Invoice Receipt' suggests a significant number of reworks or repetitions in this step, highlighting a potential area for improvement.

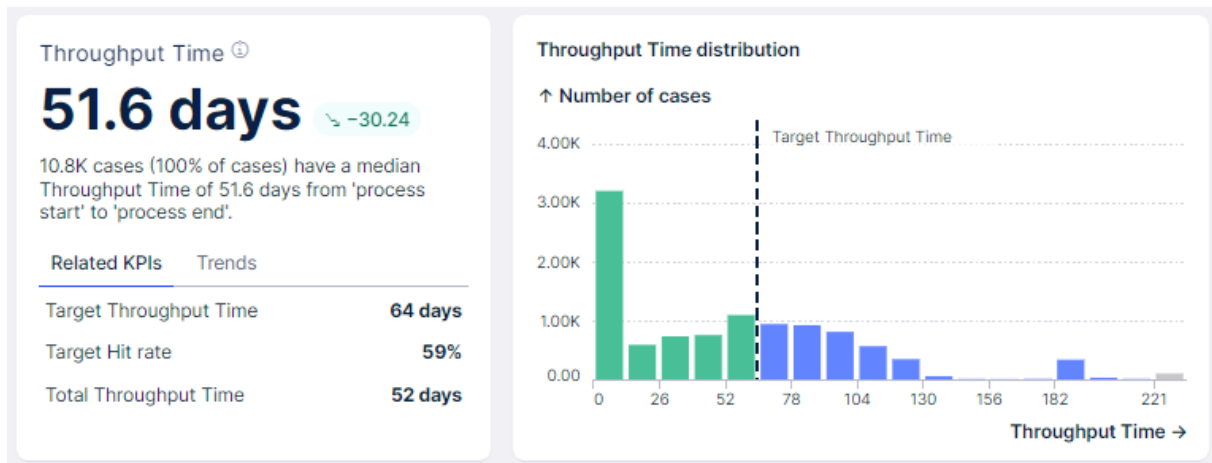
These figures illustrate the current performance and flow of the "2-way match" process. The throughput time distribution indicates that while a significant number of cases meet the target time, there are outliers that extend the overall process time. The process

explorer highlights the main activities and potential bottlenecks, particularly around the 'Create Purchase Order Item' and 'Change Approval for Purchase Order' steps. These insights can help in identifying areas for process optimization and efficiency improvements.

5.2 3-way match Invoice after GR



3 way match after GR model



Throughput Time

The following activities were excluded due to being identified as bottlenecks:

- **SRM: Awaiting approval**
- **Cancel invoice receipt**
- **Set payment block**
- **Change delivery indicator**
- **Receive order confirmation**
- **Change storage location**
- **Vendor creates debit memo**
- **Change approval for purchase order**
- **Reactivate purchase order item**
- **Record subsequent invoice**

The first figure illustrates the process explorer for the "3-way match after Goods Receipt" segment, showing the streamlined process after filtering out bottleneck activities. The key steps in the flow are:

- **Create Purchase Requisition Item:** The process begins with this step, involving 735 cases.
- **Create Purchase Order Item:** This step is crucial, involving all 10.8K cases.
- **Vendor Creates Invoice:** This step involves 6.6K cases, where the vendor issues an invoice post the purchase order.
- **Record Goods Receipt:** Another critical step with 10.3K cases, marking the receipt of goods.

- **Record Invoice Receipt:** Involving 5.69K cases, this step records the received invoice.
- **Record Service Entry Sheet:** This optional step, with 4.45K cases, records the details of the service provided.
- **Clear Invoice:** The final step, with 5.59K cases, completes the process by clearing the invoice.

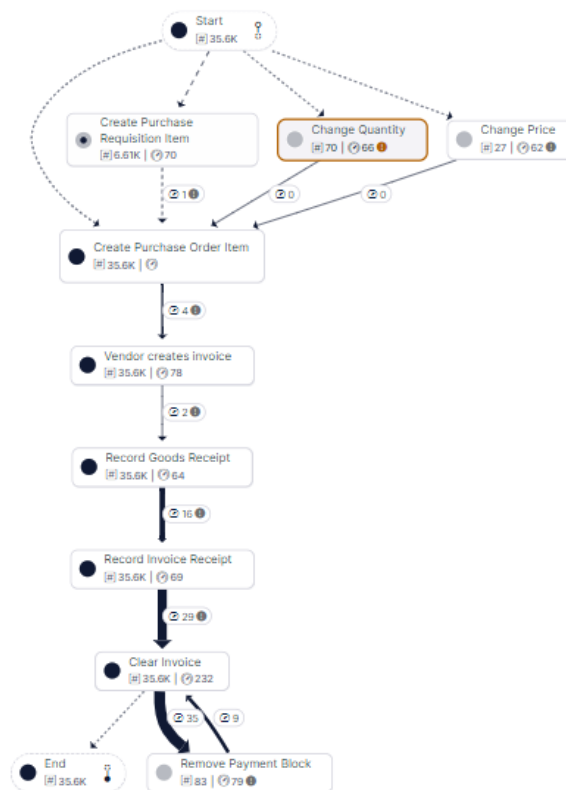
The second figure presents the throughput time distribution for the "3-way match after Goods Receipt" process, post bottleneck filtering. Key metrics and observations shows that the median throughput time is 51.6 days, The target throughput time is set at 64 days. The target hit rate is 59%, indicating that 59% of the cases meet the target throughput time.

The distribution chart shows:

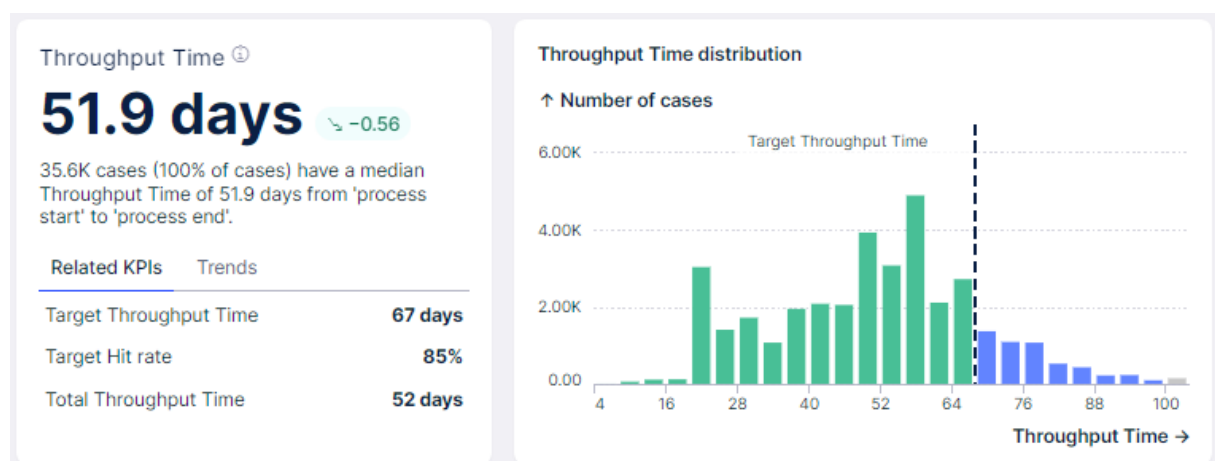
- A significant number of cases completed within the target time are represented by green bars.
- Light blue bars represent cases that met the target, while navy bars indicate those exceeding it.
- Despite significant compliance, outliers extend up to 221 days, indicating further room for process optimization.

These figures, after filtering out bottleneck activities, provide a comprehensive overview of the "3-way match after Goods Receipt" process. The process explorer outlines the main steps, focusing on critical transitions and activities without the identified bottlenecks. The throughput time distribution indicates that while a substantial portion of cases meets the target throughput time, there are outliers extending the overall process duration. These insights are essential for identifying areas for further optimization and improving process efficiency.

5.3 3-way match Invoice before GR



3 way match before GR model



Throughput Time

The provided figures give insights into the throughput times and the business process model related to the "3-way match before goods receipt (GR).

The dashboard indicates:

- Median throughput time for processing 35.6K cases is 51.9 days.
- Target throughput time is set at 67 days.
- Actual total throughput time is 52 days.
- Target hit rate is 85%.

The distribution chart shows that most cases are processed within 50-70 days, with a few outliers extending up to 100 days.

The business process model illustrates the sequence of top 9 activities from creating a purchase requisition to clearing an invoice and removing payment blocks. The key steps in the process are:

- **Create Purchase Requisition Item** 6.61K cases.
- **Change Quantity** 70 cases.
- **Change Price** 27 cases.
- **Create Purchase Order Item** 35.6k cases.
- **Vendor Creates Invoice** 35.6k cases.
- **Record Goods Receipt** 35.6k cases.
- **Record Invoice Receipt** 35.6k cases.
- **Clear Invoice** 35.6k cases.
- **Remove Payment Block** 83 cases.

The analysis based on the provided filters reveals several insights:

- The connection from **Record Invoice Receipt to Clear Invoice** has a throughput time of up to 60 days. This indicates potential delays in invoice clearance after the receipt is recorded.
- The transition from **Create Purchase Order Item to Vendor Creates Invoice** shows throughput times of up to 18 days and 15 days in two separate filters, indicating variability in the time it takes for vendors to issue an invoice after the purchase order is created.
- The step from **Record Goods Receipt to Record Invoice Receipt** has a throughput time of up to 45 days, suggesting possible delays in recording invoice receipts after goods are received.

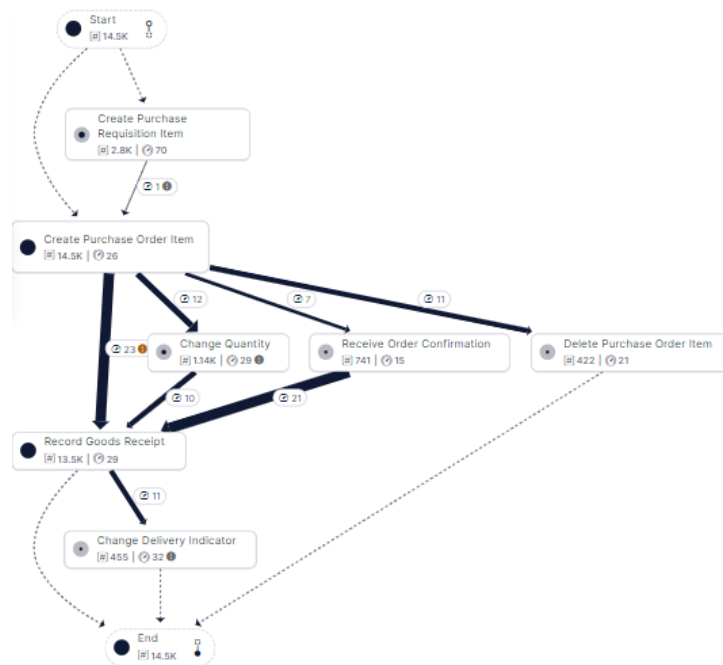
- The transition from **Vendor Creates Invoice to Record Goods Receipt** shows a relatively quick processing time of 1 to 8 days, indicating efficiency in matching invoices to goods receipts.

The analysis points to several areas for process optimization:

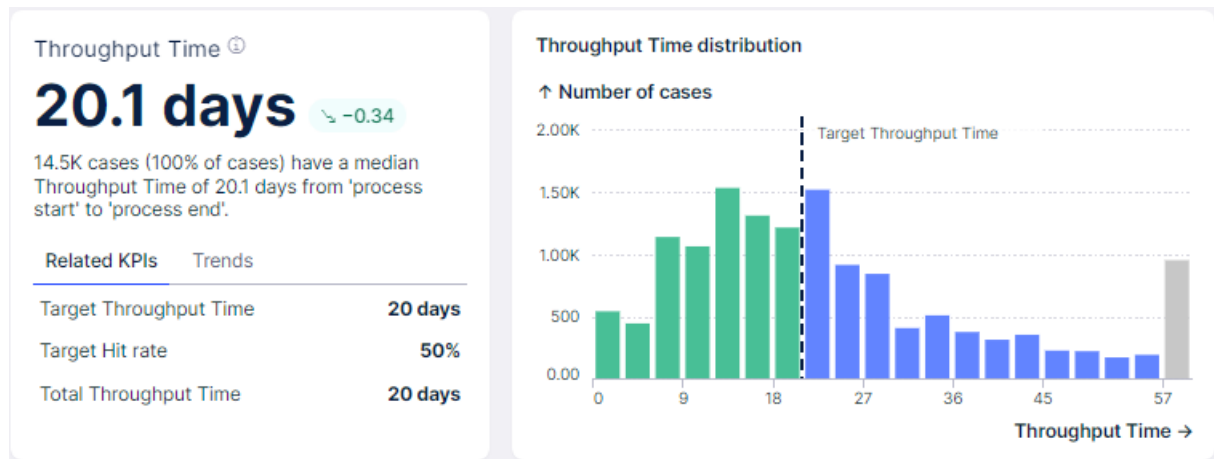
- Delays between recording invoice receipts and clearing invoices need attention. Automating parts of this process or better coordination between departments could reduce the throughput time.
- Streamlining the transition from purchase order creation to vendor invoicing can reduce variability and improve consistency.
- Identifying bottlenecks in the transition from goods receipt to invoice receipt can help shorten the overall cycle time.

The figures and the process model highlight key areas where throughput times can be reduced to meet target goals more consistently. Addressing delays in specific transitions (e.g., invoice receipt to clearing) and standardizing processes can enhance efficiency and meet business objectives.

5.4 Consignment



Consignment model



Throughput Time

The median throughput time is 20.1 days, The target hit rate is 50%, meaning half of the cases meet the target throughput time. The distribution chart shows most cases are around the 18-20 day mark, but some extend beyond the target, indicating variability. The process starts and ends with 14.5K cases, indicating a closed-loop system.

Key steps include:

- **Create Purchase Requisition Item:** 2.8K cases.
- **Create Purchase Order Item:** 14.5K cases (central step).
- **Change Quantity:** 1.13K cases.
- **Receive Order Confirmation:** 741 cases.
- **Record Goods Receipt:** 13.5K cases.
- **Change Delivery Indicator:** 455 cases.
- **Delete Purchase Order Item:** 422 cases.

The median throughput time being close to the target suggests a well-managed process with room for improvement, especially for outliers. The process flowchart visualizes key steps and highlights areas where variations occur, such as changes in quantity and deletions, which can be targeted for optimization. These insights are valuable for process improvement, reducing cycle times, and enhancing efficiency.

6 Suggestions

To address the bottlenecks identified in this BIS project, here are some detailed suggestions based on the analysis from the Celonis platform:

6.1 Automation

Automation can significantly streamline repetitive tasks, reduce human error, and speed up processes.

- **Automated Invoice Processing:** Implement robotic process automation to handle invoice matching, especially in the 2-way and 3-way match scenarios. This can reduce the time taken for manual verification and approval.
- **Automated Alerts:** Set up automated alerts for any deviations or delays in the process. For example, if a purchase order is pending approval for too long, an alert can notify the responsible personnel to take action.

6.2 Adding More Time and Resources

Sometimes, bottlenecks occur because the current resources (time, manpower, or tools) are insufficient to handle the workload.

- **Increase Staffing Levels:** If certain steps, such as 'Change Approval for Purchase Order', frequently become bottlenecks, consider adding more employees or redistributing tasks to balance the workload.
- **Extended Working Hours:** For critical periods or high-volume times, extending working hours or implementing shifts can help ensure that processes keep moving without delay.

6.3 Process Optimization

Re-evaluating and optimizing existing processes can help identify and eliminate inefficiencies.

- **Process Re-engineering:** Conduct a thorough review of the process steps to identify and eliminate any non-value-adding activities. For example, streamline the approval process to reduce the time spent in 'Change Approval for Purchase Order'.

6.4 Training and Development

Improving the skills and knowledge of the staff can lead to more efficient processes and fewer errors.

- **Regular Training Programs:** Implement regular training programs to ensure that all employees are up-to-date with the latest tools and best practices. This can be particularly useful for new tools like Celonis.
- **Knowledge Sharing:** Encourage a culture of knowledge sharing where employees can learn from each other's experiences and solutions to common problems.

6.5 Technology Upgrades

Upgrading to more advanced technology can enhance process efficiency and data accuracy.

- **Advanced Analytics:** Utilize advanced analytics and machine learning within Celonis to predict and prevent potential bottlenecks before they occur.
- **Integration of Systems:** Ensure that all relevant systems (e.g., ERP, procurement, and financial systems) are well-integrated to allow seamless data flow and reduce manual intervention.

6.6 Specific Recommendations for each Segment

- **1. 2-way Match:**

Bottleneck: 'Create Purchase Order Item' with a throughput time of 179 days.

Solution: Implement automated purchase order creation and approval workflows to speed up the process.

- **2. 3-way Match Invoice after GR:**

Bottleneck: 'Change Approval for Purchase Order' with significant reworks.

Solution: Simplify the approval process and implement automated rework identification and resolution systems.

- **3. 3-way Match Invoice before GR:**

Bottleneck: Invoice arrival before goods receipt leads to delays in matching.

Solution: Implement a pre-receipt notification system that alerts procurement and finance teams about incoming invoices and expected GRs, enabling them to prepare in advance.

Bottleneck: Delays in approval due to mismatched details.

Solution: Set up automated discrepancy reports that highlight mismatched details and route them to the appropriate team members for quick resolution.

- **4. Consignment:**

Bottleneck: Handling multiple goods receipt messages.

Solution: Use advanced filtering and automation to handle consignment processes more efficiently and ensure timely recording and matching.

By applying these strategies, we can effectively address bottlenecks, improve throughput times, and enhance overall process efficiency in this BIS project .

7 Conclusion

In this project, we analyzed our dataset using two distinct approaches to gain comprehensive insights and develop heuristic models for each data category.

7.1 Approach 1: Python and PM4Py Library

For the first approach, we used Python and the PM4Py library to meticulously explore and filter our dataset. We began with data exploration and preprocessing, handling missing values and ensuring consistency across the dataset. We then identified and addressed outliers and bottlenecks using statistical and visualization techniques. Finally, we applied various process mining models to develop heuristic models for each data category. This detailed analysis allowed us to create customized solutions tailored to our business needs.

7.2 Approach 2: Celonis Platform

For the second approach, we used the Celonis platform to streamline our analysis. Celonis facilitated quick data integration and transformation, allowing us to efficiently start our analysis. By leveraging its advanced filtering options and analytical tools, we were able to drill down into specific segments of the data and uncover targeted insights. Using Celonis' visualization and process mining capabilities, we developed and refined final models for each data category.

7.3 Addressing Bottlenecks

Using Celonis, we identified critical bottlenecks such as 'Change Approval for Purchase Order' and invoice discrepancies. Recommendations based on these insights include automation, resource optimization, process optimization, and leveraging training and technology to enhance efficiency.

These strategies aim to improve throughput times and overall process efficiency in our BIS project.

7.4 Summary

Our use of Python and PM4Py enabled detailed, customizable analysis, ensuring data consistency and addressing outliers. Celonis facilitated rapid, streamlined analysis with advanced analytics, efficiently uncovering specific insights.

By integrating these approaches, we identified critical bottlenecks such as 'Change Approval for Purchase Order' and invoice discrepancies. Recommendations for automation and process optimization aimed to streamline operations and improve efficiency, guiding strategic decisions and paving the way for continuous process improvements within our organization.

8 References

- The source code and scripts used in this analysis are available at the following GitHub repository: [BIS Repository](#).